

NAWABSHAH, Pakistan

WMO: 417490

Lat: 26.219N

Lon: 68.390E

Elev: 93

StdP: 14.65

Time Zone: 5.00 (E05)

Period: 00-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	41.4	44.5	33.8	28.5	58.8	36.0	31.2	62.0	13.4	65.8	11.8	67.9	1.9	0	0.436

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	26.8	116.2	77.8	113.1	77.9	110.8	78.1	85.1	102.2	84.1	101.8	83.4	101.2	7.9	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
80.9	161.2	93.9	80.1	156.7	93.7	78.9	150.7	93.4	49.8	102.5	48.5	102.5	47.6	101.3	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.0	14.4	12.8	DB	38.4	120.3	4.4	1.5	35.3	121.3	32.7	122.2	30.2	123.1	27.0	124.2	(4)
(5)			WB	36.5	87.2	4.2	1.3	33.5	88.1	31.0	88.9	28.7	89.6	25.6	90.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	81.9	60.3	66.9	77.9	88.3	95.7	96.5	94.1	91.2	90.0	84.5	73.3	63.2	(6)
(7)		DBStd	13.16	3.63	5.39	5.77	4.51	3.40	2.95	2.91	2.60	2.23	3.92	4.72	4.26	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	277	152	37	2	0	0	0	0	0	0	0	1	85	(9)
(10)		CDD50	11639	318	473	864	1150	1418	1395	1366	1277	1200	1069	699	410	(10)
(11)		CDD65	6442	5	90	401	700	953	945	901	812	750	604	250	31	(11)
(12)		CDH74	98017	263	1324	5403	10741	15864	15633	14253	12099	10884	8006	2973	574	(12)
(13)		CDH80	64832	27	474	3246	7382	11557	11334	9802	7679	6805	5008	1413	105	(13)
(14)	Wind	WSAvg	5.9	3.8	4.3	4.6	5.6	7.0	9.4	9.5	8.4	7.1	4.4	3.1	3.4	(14)
(15)	Precipitation	PrecAvg	4.6	0.0	0.1	0.1	0.1	0.0	0.4	1.5	1.5	0.4	0.0	0.1	0.2	(15)
(16)		PrecMax	16.3	0.4	0.3	1.3	1.5	0.4	4.3	7.8	5.3	2.7	0.2	1.0	1.1	(16)
(17)		PrecMin	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	4.3	0.1	0.1	0.3	0.3	0.1	1.0	1.7	1.9	0.7	0.0	0.2	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	82.8	93.5	107.8	115.1	120.1	118.6	111.3	107.3	107.6	106.2	98.4	86.3	(19)
(20)			MCWB	62.9	67.6	71.7	75.1	77.7	79.7	82.7	82.2	79.5	75.5	70.8	65.7	(20)
(21)		2%	DB	79.2	89.7	103.9	111.5	116.8	114.9	109.0	104.4	105.1	104.1	94.8	82.7	(21)
(22)			MCWB	61.0	67.0	71.0	74.3	77.5	79.4	82.4	82.2	80.0	75.7	70.5	63.7	(22)
(23)		5%	DB	77.0	85.9	99.4	109.1	114.6	111.6	106.7	102.9	102.6	102.0	91.5	80.2	(23)
(24)			MCWB	60.0	65.2	70.2	74.0	77.4	80.3	82.4	81.9	79.9	75.0	69.2	62.3	(24)
(25)		10%	DB	74.0	82.2	95.3	105.8	111.6	109.3	104.5	100.8	100.8	99.0	88.1	77.0	(25)
(26)			MCWB	58.2	63.2	69.3	73.2	77.3	80.5	82.4	81.6	79.7	74.4	67.9	60.3	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	65.4	71.2	76.9	80.9	84.3	85.5	86.6	85.5	84.8	81.4	74.9	67.9	(27)
(28)			MCD B	79.3	87.6	97.2	107.3	109.3	105.1	102.7	100.3	100.6	98.9	90.7	79.1	(28)
(29)		2%	WB	62.9	68.6	74.0	78.5	82.5	84.4	85.2	84.4	83.3	79.6	72.7	65.6	(29)
(30)			MCD B	77.1	85.9	96.2	104.4	107.8	104.6	102.0	99.1	99.0	97.0	89.4	79.5	(30)
(31)		5%	WB	61.0	66.5	72.0	76.5	81.4	83.5	84.2	83.5	82.0	78.1	70.9	63.7	(31)
(32)			MCD B	74.7	83.4	94.6	102.1	105.8	104.0	101.8	98.3	97.3	95.4	87.0	77.6	(32)
(33)		10%	WB	59.3	64.4	70.2	74.8	80.1	82.6	83.3	82.6	81.0	76.9	69.1	61.8	(33)
(34)			MCD B	72.2	79.9	92.1	100.4	103.9	103.5	100.8	97.6	96.4	93.8	85.0	75.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	28.9	30.3	33.0	34.1	33.1	26.8	22.0	20.6	23.9	30.6	30.3	28.6	(35)
(36)			MCDBR	32.6	34.8	38.8	38.7	36.3	31.1	24.8	23.2	26.4	33.0	33.1	31.1	(36)
(37)			MCWBR	16.5	15.4	13.4	11.8	9.7	8.6	7.9	8.0	8.8	10.3	13.2	14.5	(37)
(38)		5% WB	MCD BR	30.7	31.7	35.2	35.8	32.3	25.8	22.0	20.7	22.9	28.3	30.4	29.1	(38)
(39)			MCWBR	16.7	15.1	14.7	13.7	10.5	8.6	8.2	8.2	8.6	10.4	13.4	14.8	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.472	0.480	0.539	0.607	0.649	0.717	0.784	0.717	0.612	0.562	0.565	0.527	(40)
(41)		taud		1.914	1.933	1.805	1.698	1.654	1.570	1.504	1.606	1.761	1.772	1.726	1.786	(41)
(42)		Ebn at Noon		239	250	243	230	221	205	191	203	221	223	208	213	(42)
(43)		Edh at Noon		54	56	68	77	81	87	93	84	70	66	64	59	(43)
(44)	All-Sky Solar Radiation	RadAvg		1275	1556	1886	2113	2243	2145	1902	1876	1863	1693	1348	1208	(44)
(45)		RadStd		60	71	70	56	36	60	115	86	94	69	59	56	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	-0.86	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page